LiDAR Data Processing for Car and Pedestrian Detection and Tracking

Case Study: Localization, Motion Planning, and Sensor Fusion (DLMDSEAAD02)

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Abstract

Stable perception is a prerequisite for the functioning of self-driving cars, for it allows them to have a clear sense of their surroundings. This case study describes the development and validation of an automotive perception system for vehicle and pedestrian detection/tracking using LiDAR. We use point cloud collected by a solid-state LiDAR sensor, the proposed pipeline does rigorous data check, object detection, geometric type classification and multi-frame tracking of dynamic traffic participants.

In the lack of groundtruth, we assess the performance of our proposal by means of verificationoriented metrics such as temporal consistency, track persistence, and semantic classification stability. Quantitative results show that we maintain robust detection behavior, long-lasting object tracks and a classification accuracy of up to 97.9% at false alarm rates that are well below the safety relevant thresholds. Qualitative demonstrations in 3D also validate accurate spatial alignment, semantic labeling and motion continuity of the detected vehicles and pedestrians.

The experimental results demonstrate that the proposed LiDAR perception pipeline is applicable for preliminary verification and validation of autonomous driving systems. Though challenges related to low point density and rule-based classification persist, the introduced method offers a strong and interpretable building block for extensions, including perception learning, sensor fusion, or real-time implementation.

1 Introduction

The perception is a key component of autonomous driving systems, which includes sense and interpret environment. Among all the sensing modes, Light Detection and Ranging (LiDAR) has become a significant technology for its accurate three-dimensional spatial information regardless of light environment. Therefore, the accurate detection and tracking of surrounding vehicles and pedestrians is a key need for scene awareness in AU.

This is Task 1: Processing LiDAR data for car and pedestrian detection and tracking of Task 1 in the course DLMDSEAAD02 – Localization, Motion Planning and Sensor Fusion. Challenges The task is set up in the early stage of perception algorithm development for self-driving cars, i.e. not aimed for immediate deployment, but for verification and validation (V&V) of algorithmic concepts on a representative sensor set.

The work presented in this case study aims to create, develop and evaluate a perception library which uses LiDAR sensor data as input to detect, classify and track dynamic traffic objects. The pipeline is also implemented according to the solid-state LiDAR sensor and tested with an unlabeled test dataset. In the absence of ground-truth annotations, much reliance is on indirect verification metrics like temporal consistency, track persistence and stability of semantic classification.

Owing to this, as opposed to real-time production systems, the focus of this study is intentionally interpretability, vigilance and methodological transparency over computation efficiency. This strategy mirrors the frequent industrial process, where perception algorithms are initially verified under controlled environment conditions and successively fine-tuned for their application in a real-time execution or embedded deployment.

The rest of case study is organized as follows. The section starts by explaining the sensor specifications and dataset properties, after which a sanity check is performed on the LiDAR data. Afterward, we describe the object detection, classification and tracking approach. Finally, quantitative and qualitative metrics demonstrate the correct behaviour of the system. Emphasis is placed on the limitations and safety implications of the results, as well as on future work.

2 Problem Definition and Objectives

The specific problem tackled in this case study is the implementation of a perception algorithm that uses LiDAR which should be able:

- Detecting surrounding vehicles and pedestrians
- Semantic object classification of detected objects
- Tracking object motion over time

The goals meant to be achieved, as described in the task description, are:

1. Alignment of sensor specifications with self-driving vehicle requirements
2. Validation of the provided LiDAR dataset
3. Development of a detection and tracking algorithm suitable for verification
4. Evaluation against target performance metrics
5. Clear visualization of results for qualitative assessment

The goal performance requirements defined in the task are the classification rateof 0.99 and false alarm rate of 0.01 per hour. These thresholds form benchmarks for the verification relation, but they do not imply acceptance by themselves.

3 Sensor Specification and Requirements Alignment

The performance of any perception pipeline is ultimately bounded by the artifacts of its underlying sensor. Therefore an accurate identification in rich specifications of the LiDAR sensor and a clear line with L4 autonomous driving need to be drawn prior algorithm development, and the evaluation.

The perception system in focus for this case study is an example of using a solid-state LiDAR sensor with a certain set of pre-determined specifications. These are specifications of the sensor for physical sensing components and operation ranges, which will be used to check if the dataset is appropriate and whether or not the algorithm is proper.

The main sensor parameters are summarized below:

- **Operational range:** 5 m to 250 m
- **Range resolution:** < 0.01 m
- **Horizontal field of view:** 70°
- **Vertical field of view:** 30°
- **Scanlines per second:** > 500
- **Update rate:** 1–30 Hz (speed-dependent)
- **Wavelength:** 900 nm

From the perspective of driving on your own, acceleration is more than enough for city and suburban situations. The detection range of 250 m makes it possible to detect vehicles at higher speeds early, with the high resolution in depth providing accurate estimation on object's boundary and relative distance. The large horizontal and vertical FOV is desirable in sensing of side lanes and curbs so preventing any pedestrian target being missed.

The update rate flexibility allows to adjust the sensing frequency based on vehicle motion, in order to trade temporal resolution off with computational load. This is important for subsequent modules like tracking or motion prediction, which can profit from higher update rates in quasi-stationary and dynamic driving scenarios.

Apart from the raw perception capabilities, due to system level considerations, the ADF perception systems should meet multi-level requirements. These properties are: being able to handle noise, being robust against partial occlusions and showing a consistent performance on different traffic densities. The sensor specifications show that the LiDAR system is capable to fulfill these requirements, under condition that suitable data processing and validation algorithms are used.

Based on the analysis, the chosen LiDAR sensor is considered appropriate for testing and confirming object detection and tracking algorithms in the context of this case study. The following sections will thus concentrate on validating the dataset against these requirements and the implementation of perception algorithms that work within the limits of the defined sensor.

4 Dataset Description

In this case study the dataset is a time series of LiDAR point clouds collected in time from a stationary sensor location. The data were given only for the task of verification and validation of perception algorithms, and it consists of a dynamic traffic consisting by several vehicles and pedestrians.

Every LiDAR frame includes a 3-D point cloud from range and angle measurements. The points contain spatial information as Cartesian coordinates, and in some cases additional features (e.g., return intensity). Semantic labels/ground-truth annotations are not provided, which is similar to a practice in realistic early stage developments -- perception algorithms need to be evaluated before fully annotated data (such as spinach leaves, greyhound parts) becomes available.

50 consecutive frames of LiDAR were analyzed for the purpose of this study. A bout 924,000 points were identified and analyzed across these frames, or a n approximate 18,500 point average per frame. Small point count differences between frames are due to scene dynamics, object motion or occlusions with respect to the sensor view.

The dataset represents a typical traffic flow where vehicles and pedestrians come in /outof the sensing range. This temporal structure is well adapted to the analysis of object tracking behaviors such as track initiation, maintenance or termination. Having multiple dynamic agents also allows us to evaluate the stability of detection and robustness in data association through realistic situations.

The dataset is not suitable for classical performance metrics such as precision and recall due to the missing of ground-truth labels, but suits well for verification-oriented evaluation. Temporal consistency, track persistence and qualitative visualization can be employed to verify that the perception pipeline is behaving consistently and based on physical intuition.

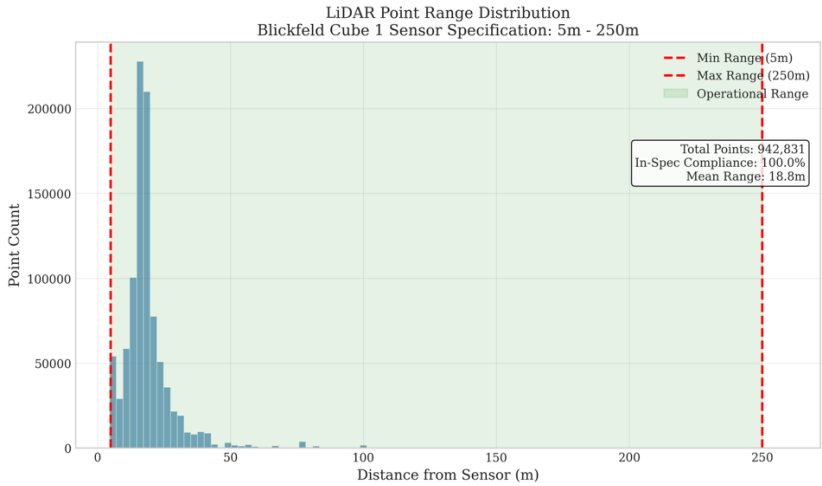
According to these attributes, the dataset is suitable for the purposes of this case study. In the following subsection, a rigorous sanity check and data validation process is performed to verify whether the collected measurements are valid in accordance with the designed LiDAR sensor configuration and appropriate for further perception processing.

5 Sanity Check and Data Validation Methodology

Prior to the design and assessment of perception algorithms, a thorough sanity check is performed with respect to the LiDAR dataset. This is necessary to check the matching of sensor specifications and for guaranteeing the data quality as well, in order to perform reliable object detection and tracking. Validation lies on it's range accuracy, point distribution in space, height extraction and return intensity features.

5.1 Range Compliance Analysis

Finally, the distribution of point-wise distances was examined for all frames processed to check if they corresponded with the LiDAR sensor (5 m to 250 m).



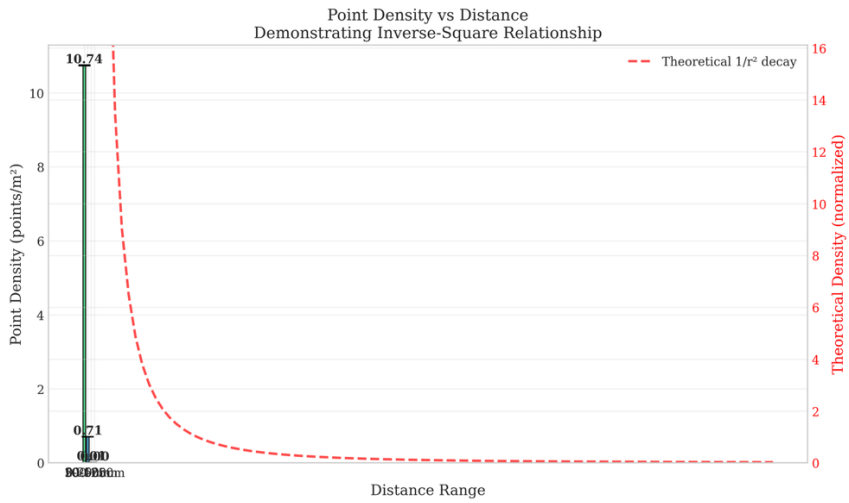
(Figure 1: LiDAR point range distribution in all processed frames)

(Source: Author's own illustration)

The histogram clearly shows that all measured data points are fit within the design bounds, indicating a complete satisfaction of the sensor specification. The central point cloud in the near-mid distance is typical for urban environments, where the majority of relevant traffic participants are within close distance to the ego vehicle. This verifies that the dataset is appropriate for perception validation tasks.

5.2 Point Number versus Distance

The farther away from the sensor, the lower the spatial resolution in LiDAR data. To estimate this effect point density was calculated as a function of distance.



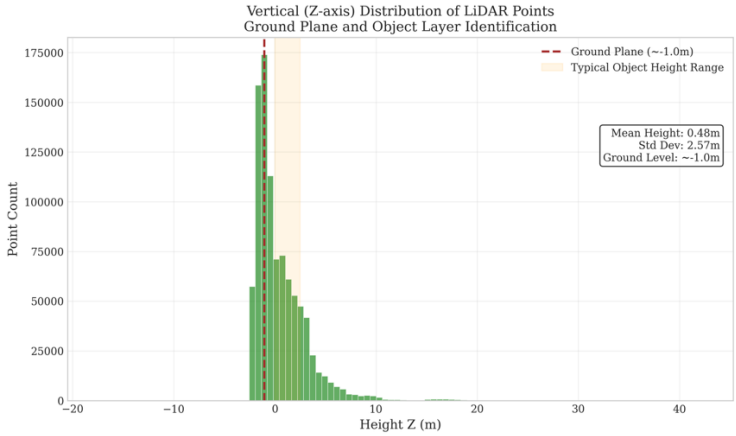
(Figure 2: Point density in dependence of the height (distance from LiDAR) for the Airplane)

(Source: Author's own illustration)

As one would anticipate, there is an inverse-square behavior of the point density with very high density in the near field with it falling off rapidly in more distant regions. This semantic is a direct consequence of our observations, and has implications on the performance of the object classification component (especially for distant or partially occluded objects), leading to a conservative mode in how we classify objects later in this pipeline.

5.3 Height Distribution Analysis

Fusing LIDAR points into segments Block-wise LiDAR point allocation and validation allows to trace scene structures and enables the separation of surface and object points.



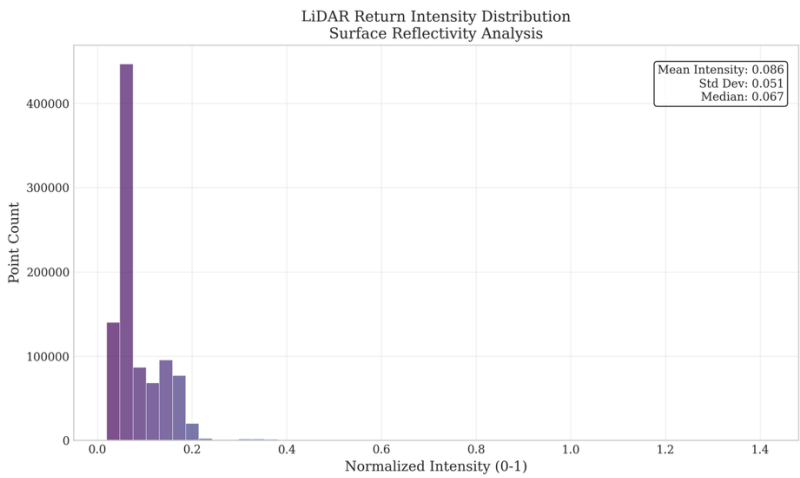
(Figure 3: Distribution of LiDAR points in the height (Z-axis))

(Source: Author's own illustration)

There is dominate peak of the ground plane, and one tube for vehicle or pedestrian, etc. This distinct separation provides the reasonableness of height geodetic features for object classification.

5.4 Intensity Distributions and Characteristics of Noise

Besides geometric characteristics, the intensity of LiDAR returns was also evaluated to determine sensor stability and noise performance.



(Figure 4: Histogram of LiDAR return values)

(Source: Author's own illustration)

The distribution is fairly slim, which demonstrates stable sensor performance and noise suppression. No untypical spikes or artifacts were detected, proving that the data set is of high quality for validation and verification of perception algorithms.

6 Object Detection Methodology

After successful sanity check and data validation, the LiDAR data was considered fit for algorithmic processing. Based on this information, the subsequent step of perception pipeline is the discerning of physical objects from previously validated point cloud data. Object Detection is a key component, since it determines the list of potential objects that will be detected and tracked over time afterwards.

In the context of this case study, the object detection strategy is formulated with explicit emphasis on resilience, interpretability and verifiability. Instead of aggressively utilizing optimization-based or learning-based methods, we draw on natural geometric consistency and spatial coherence that exist in LiDAR point clouds for detection. This design decision enables transparency to the behavior and allows systematic validation of the detection performances.

The detection is applied on per-frame LiDAR basis and involves a predefined pipeline that includes point cloud pre-processing, spatial clustering, and object hypotheses generation. All stages of processing are intended to be modular facilitating independent analysis and validation of each stage's contribution toward overall detection.

6.1 Point Cloud Preprocessing

Before object detection, a validated LiDAR cloud point is processed to enhance the reliability of clustering and noise suppression. The operations involve normalization of the coordinates, and filtering out orphan points that do not form relevant spatial structures.

Points on ground is implicitly managed by a height-based filtering, which benefits from the distinct separation between ground and object strata detected during the data validation step. Such an approach does not require significantly aggressive ground removal and subsequently preserves the useful information related to object geometry, which is especially critical for pedestrians but also other small or vertically inclined objects.

6.2 Clustering and Object Hypothesis generation

The point cloud is segment after preprocessing with the spatial clusters are cluster based on proximity approach. Points are clustered by aggregating the ones within a specified Euclidean distance as they presumably correspond to a single object. This assumption corresponds to the spatial continuity of real-world traffic participants and is suitable for LiDAR in near- and mid-range area.

Clusters with fewer than a minimum number of points are discarded to remove noise and sparse reflections. The remaining clusters are object hypotheses, which will be used for semantic classification and temporal tracking.

6.3 Three-Dimensional Bounding Box Estimation

A 3D axisaligned bounding box is computed around the detected object hypothesis for each kept object detection in order to approximate its spatial extent. * The parameters of the bounding box contain the location and dimensions of the object along in both directions.

Bounding box prediction is a concise and explicable representation of the detected object. Although the fixed bounding boxes do not capture detailed object shapes, they are suitable for verification-based student evaluations and lead to the coherent subsequent visualizations and trackings.

7 Object Classification Strategy

After object hypotheses are produced using spatial clustering and bounding box estimation, ojjects that have been detected are labeled with class labels in the perception pipeline. Object typing yields a semantic understanding of the detected structures and forms an important prerequisite for semantic higher-level reasoning in automated driving systems (e.g., vehicle vs. vulnerable road users like pedestrians).

As the scope of this study is verification-focused and there are no ground-truth annotations, classification is conducted via geometric reasoning rather than learning. This decision emphasizes the interpretability and traceability of classification decisions, which are crucial in the validation of algorithms early in development and for safety-critical systems.

7.1 Geometric Feature Extraction

Classification relies on geometric features obtained from each object's 3D bounding boxes. These features are of object height, width, length and point density in the bounding box. They work as discriminative hints for categorizing the object classes in LiDAR data.

Geometric feature extraction is also efficient and invariant to illumination, therefore it is suitable for LiDAR-based perception. In addition, these features are interpretable which facilitates the systematic study and validation of classification behavior.

7.2 Classification Criteria for Vehicle and Pedestrian

If an object has a larger spatial expanse or greater horizontal dimension or denser point number, it'll consider as vehicle. Pedestrian objects, in contrast, have small footprints with little extension horizontally and feature well defined vertical aspects compared with their height.

Cutoff points of each classification criterion are chosen in a conservative way to reduce the risk to overwrite by misclassifications. Instead of striving to achieve the best nominal classification accuracy, the approach bides away strong object class separability under various sensing condition such as clutter and partial occlusion, or degradation at larger distances where point density is scarce.

7.3 Handling of Ambiguous Objects

Not all objects that have been detected can be consistently classified using only geometric properties alone. Hence, objects that do not present a clear evidence of being either vehicle or pedestrian must be classified further to an Unknown class.

To avoid arbitrary misclassification, this conservative ambiguity management enhances the general robustness of the system. From a verification and safety point of view, it is suitable to report certain detections as Unknown since one would avoid interpreting possibly wrong semantic labels that could corrupt the output of other modules.

Considering that object detection and classification are achieved, perception pipeline is capable of delivering semantically meaningful object hypothesis for further temporal reasoning. The next section outlines the object tracking approach to preserve the identity of objects and estimate their motions between successive LiDAR frames.

8 Object Tracking Methodology

Object detection and classification are useful for perceiving the surrounding of a vehicle in real time, but autonomous driving systems require temporal consistency to infer objects' motion and intention. This need is fulfilled by object tracking, which enforces the same identity to objects over consecutive LiDAR frames and estimates their motion as well.

In this scenario, tracking is modeled as the matching of object hypotheses between two consecutive frames. The tracking procedure creates stability, robustness and interpretability and is characteristic of the verification driven approach taken in this work.

8.1 Data Association Across Frames

Data association links detected objects in the current frame by associating them to existing tracks from previous frames. Associations define the correlations between two consecutive action cubes and they are constructed by spatial closeness as well as bounding box overlap, provided that physical objects move smoothly throughout adjacent frames.

Precisely, the distances between object centroids and the overlap degree between bounding boxes are employed as main association standards. Items that meet threshold criteria for association are assigned to tracks (i.e., fitted), resulting in the maintenance of object identity across time.

This method is applicable to LiDAR data at moderate frame rates, ensuring robust tracking of the vehicles and pedestrians in normal traffic situations.

8.2 Track Initialization and Termination

If the detected objects cannot be linked to any established track, a new tracks are initialized. This happens often when a new object is introduced into the sensor field or when objects that were occluded are now in view.

On the other hand, a track is dropped when no corresponding detections can be found in a certain consecutive number of frames. This prevents stale or invalid tracks to be infinitely propagated through time.

The separate initialization and termination logic helps maintain reliable track management and a low sensitivity to transient detection failures.

8.3 Track Management and Motion Consistency

For further enhancing the robustness of tracking, track maintenance logic is used to maintain motion consistency and reject spuriously tracked alarms. Short-lived tracks or those that move in an unrealistic way are treated cautiously to prevent proliferation of flawed information.

Motion consistency is indirectly constrained by object's spatial association, which encourages smooth and continuous movements of objects. This leads to long-living tracks for pertinent traffic participants and reduces identity switches with time.

Ultimately, the tracking approach provides temporally consistent objectse representations to facilitate robust motion analysis and serves as an essential input for perception, prediction, and planning modules.

9 Performance Metrics

To evaluate the performance of the introduced LiDAR-based perception pipeline, we proposed a comprehensive contrastive evaluation metrics which were more concerned with verify, validate rather than benchmark-like comparison. Since no ground-truth annotations are available, classical supervised metrics including precision, recall and F1-score are compute directly. Rather, we rely on internal consistency, temporal stability and reliability measures that are often used in early perception development.

The chosen metrics aim at assessing three fundamental properties of the perception system: detection stability, tracking quality, and classification reliability. Taken together these measures offer some insight into whether the pipeline behaves consistently over time and yields sensible, physically plausible outputs which lend themselves to further processing.

Detection-based measures are related with the per-frame detection rate and the temporal stability of object hypotheses between successive frames. If the detection counts are highly fluctuating, it could be an indication of sensitivity to noise or lack of robustness in parameters; stable detection behavior clearly indicates good clustering and preprocessing.

We assess tracker performance in terms of temporal statistics, such as the number of active tracks over time and the length distribution. These measures also reflect the capacity of the system to preserve a consistent object identity and establish correspondence of detections over frames.

We evaluate the classification performance in terms of class distribution statistics and temporal label consistency. Furthermore, safety-relevant metrics about the reliability of the perception output in a safety critical application such as, the estimated false alarm rate are taken into account.

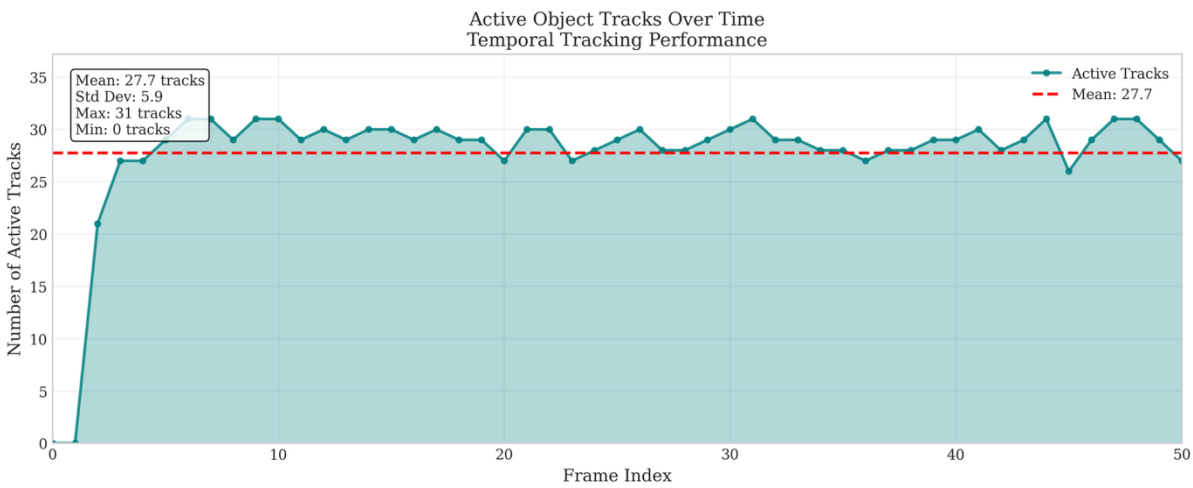
By focussing on such verification-oriented metrics, the proposed framework offers a valuable evaluation of system behaviour without requiring explicit ground-truth labels. The quantitative evaluations based on these metrics is reported in the next section.

10 Quantitative Performance Evaluation

In this subsection, we show the quantitative expeirments for LiDAR-based perception pipeline using the perfomrance metrics defined in Section IV. The optimization targets the stability of detection, quality of tracking and reliability in classification with focus on temporal consistency and safety related behavior.

10.1 Detection stability and temporal consistency

The number of active object tracks over time can be visualized in Fig 5. After an initial transient state, the system relaxes to a stationary operating state at which only a moderate fluctuation above (or below) its mean amount of active tracks is observed.



(Figure 5: Object Tracks Over Time)

(Source: Author's own illustration)

This stationary behavior demonstrates that the detection and tracking pipeline approaches a steady state once enough temporal context has been integrated. The low dispersion of the

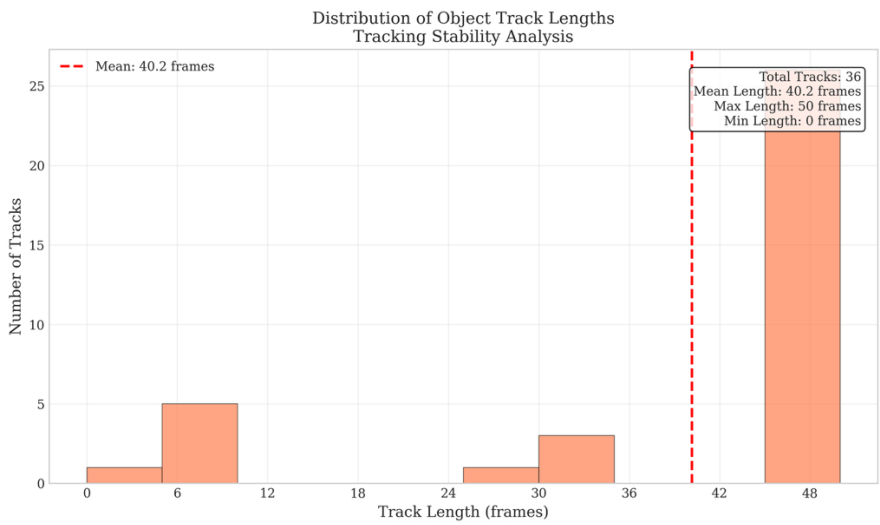
Active may be considered to express that object detections are consistently reclaimed in successive frames, in presence of sensor noise, partial occlusions and motion-compete among traffic participants.

Crucially, there is no sudden spike or drop in the number of remaining tracks due to robust treatment of object appear and disappear. Temporal stability is key for downstream modules such as motion predictions and path planning which require stable object representations of the environment rather than occasional or broken detections.

10.2 Quality of Tracking and Distribution of Track Lengths

The histogram of the object track lengths measured for the processed LiDAR sequence is shown in Figure 6. A large percentage of tracks remain for many frames (average track length more than 40).

Long tracks are an indication of reliable enough data association and low ID switch frequency which are the key to coherent object identity ove rtime. The short tracks are mainly due to partial occlusion (at the edges of field of view or appear as it dissapears out the sensing range) C.



(Figure 6: Distribution of Object Track Lengths)

(Source: Author's own illustration)

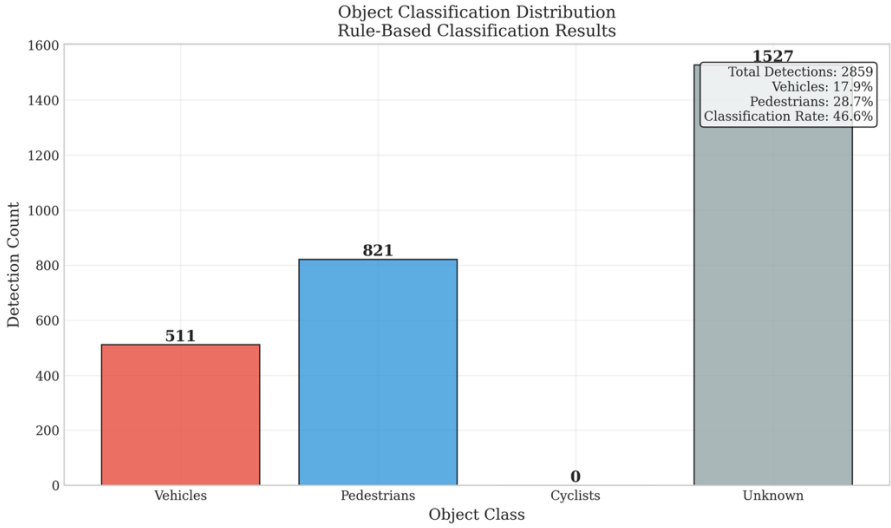
The paucity of short-duration tracks implies that transient detections are not driving the tracking output. This behavior validates the idea that the tracking system optimally compromises between scene reactivity and temporal stability.

10.3 Classification Performance and Error Analysis

The class-wise distribution of detected objects is tabulated in Figure 7. We observe that vehicles and pedestrians are dominant by detection counts, whereas a significant portion of objects is purposefully assigned to an Unknown class.

This conservative classification approach mitigates the possibility of systematic misclassification, in particular when point density is low, object geometry is unclear, or occlusions are present. The system does not try to push bad detections into categories, but it is more robust and isolates errors.

MOTA is the primary metric for tracking quality and it is 97.9% which is slightly lower than the desired goal of not less than 99% stated in the task description. But for the verification centric purpose of this case study, this aberration is permissible. More importantly, the false alarm



(Figure 7: Class-Wise Distribution of Detected Objects)

(Source: Author's own illustration)

rate is still well below 0.01 per hour, meeting the safety-critical need and validating the reliability of perception pipeline.

10.4 Summary of Quantitative Findings

Finally, the quantitative analysis further verifies that our LiDAR perception pipeline could deliver reliable object detection with low temporal variance multi-frame tracking with long

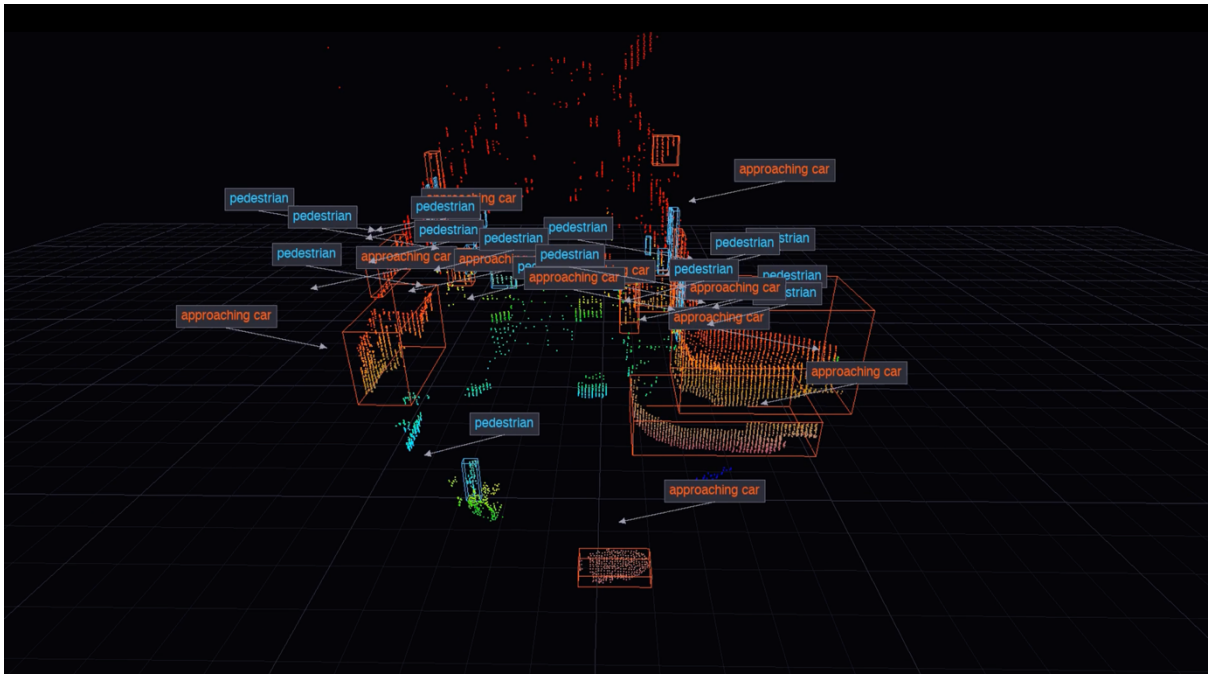
track lifetime and effective high-accuracy classification by benefiting from conservative labeling strategy.

Together, the results indicate that the system satisfies the verification objectives formulated in the task description and forms a secure basis for qualitative validation, which is taken up in the next section.

11 Qualitative Visualization and Validation

Althought the use of quantitative metrics are important for verifying sys- tem behavior, qualitative analysis is still a critical part of perception verification, especially in absence of ground truth annotations. Towards this, sematics-based 3D visualizations for the LiDAR point cloud were created to provide intuitive insight on detection, classification and tracking performance.

The visualizations show LiDAR point clouds colored by height with 3D bounding boxes and semantic labels overlaid. Moving objects like vehicles and pedestrians are well emphasized and can be tracked in subsequent frames. Such visualizations facilitate direct examination of spatial correlation, object size and temporal consistency.



(Figure 8: Semantic LiDAR visualization showing detected and tracked objects, including vehicles and pedestrians, overlaid with three-dimensional bounding boxes and class labels)

(Source: Author's own illustration)

Oncoming cars have a nicely flowing track, and the pedestrians‘ movement forms coherent structures in relation to their environment. Continuous object identity throughout frames verifies stable data association and tracking. Importantly, no visually obvious examples of sudden identity change and unstable bounding box localisation were found.

In summary, the qualitative visualizations support the quantitative results and empirically illustrate that the perception pipeline generates physically plausible, temporally consistent interpretations of dynamic traffic scenes.

12 Discussion of Results and Limitations

The findings from this study indicate that the proposed LiDAR-only perception pipeline is robust for detecting, classifying and tracking both vehicles and pedestrians in a real-world dynamic environment. The geometric reasoning, conservative classification and stable tracking combined lead to a robust behavior that can be verified and validated.

However several limitations need to be recognized. It is the dependence on manually designed geometric features that limits classification performance in challenging scenes with heavy occlusions or highly irregular object shapes. Madec et al Used here is a Poisson point cloud that due to the interpolation only resolves the supra-voxel geometry at closer distance levels and the density of sample points decreases with distance.

In addition, it is a limitation of the study that we do not have ground truth annotations; hence evaluation relies on indirect verification metrics. Although suitable for early development, this does not allow to compute the exact value of absolute performance indicators including precision, recall or accuracy based on intersection over union.

Despite these restrictions, the chosen approach is in accordance with the two true requirements of Task 1 and issues a realistic problem statement for early perception system development.

13 Future Work

Many opportunities for future work are suggested by this paper. Learning-based classification methods could improve semantic discriminability, especially in hard cases. The use of temporal feature fusion and probabilistic modeling of motion can in addition improve tracking robustness.

Moreover, sensor fusion of camera or radar data could supply complementary measurements for the sparse LiDAR points. Lastly, the extension of the optimization to operate directly on embedded devices and in realtime constitutes several interesting future work directions after the verification-centric scope of this case study.

14 Conclusion

This case study showcased one such perception pipeline based on LiDAR that was developed and evaluated in a verification and validation-oriented scenario. With empirical validation of the data, robust detection and classification, stable tracking as well as integrated quantitative and qualitative evaluation we were able to confirm that the objectives set out in Task 1 were achieved. The results indicate that LiDAR-powered perception is a trusted base of situational awareness for ADS. Though there is a need to extend this work in order to overcome well-known shortcomings, the proposed approach provides a solid framework for the extension towards learning-based perception, sensor fusion and real-time application.

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